| | Fall Protection Procedure | Revision: #3 |
|---|---|---|
| | 75D-04.0-11 | Effective: 8/13/20 |

## 1.0 Purpose:

Falls are among the most common causes of serious work-related injuries and deaths in the work environment. This procedure establishes the minimum requirements and methods necessary to protect workers from potential fall hazards at the DCF in accordance with OSHA standards. This procedure outlines approved fall protection systems and the roles and responsibilities involved to safely perform work from heights greater than 4ft and from controlled access areas where possible fall hazards exist.

## 2.0 Scope:

The fall protection program applies to all DCF employees and contractors and covers fall protection responsibilities, acceptable practices, and required procedures.

## 3.0 Definitions:

3.1 Anchorage – a secure point of attachment for lifelines, lanyards, or deceleration devices. (Anchorage or Anchor point is also known as the tie-off point)

3.2 Buckle – any device for holding the full body harness closed around the employee's body.

3.3 Connector – a device used to connect parts of the personal fall arrest system and positioning device systems together (i.e., carabiners, buckles, D-rings, or snap-hooks).

3.4 Controlled Access Zone – areas without appropriate fall protection and where access is controlled.

3.5 Competent Person – a person who, because of training and experience, is capable of identifying hazardous or dangerous conditions associated with falls and is authorized to take prompt corrective measures to eliminate the conditions or hazard.

3.6 Deceleration Device – any mechanism, such as a rope grab, rip stitch lanyard, specially woven lanyard, tearing or deforming lanyards, automatic self-retracting lifelines/lanyards, etc., which serves to dissipate a substantial amount of energy during a fall arrest, or otherwise limit the energy imposed on an employee during fall arrest.

3.7 Energy Absorbing Lanyard (EAL) - a type of safety lanyard that lessens fall forces by ripping stitched webbing. (Energy absorbing lanyard is also known as a shock absorbing lanyard.)

3.8 Deceleration Distance – the additional vertical distance a falling employee travels, excluding lifeline elongation and free fall distance, before stopping, from the point at which the deceleration device begins to operate. Measured as distance between employee's full body harness attachment point at the moment of activation (or onset of fall arrest forces) of the deceleration device during a fall, and the location of that attachment point after the employee comes to a full stop.

3.9 Failure – load refusal, breakage, or separation of component parts. Load refusal is the point where the ultimate strength is exceeded.

3.10 Free Fall – the act of falling before a personal fall arrest system begins to apply force to arrest the fall.

3.11 Full Body Harness – straps which may be secured about the employee in a manner that will distribute the fall arrest forces over at least the thighs, pelvis, waist, chest and shoulders with means for attaching it to other components of a personal fall arrest system.

3.12 Guardrail System – a barrier guard consisting of a top-rail, mid-rail, and toe-board for all open sides of a walking/working surface (i.e., roof, floor, platform, scaffold, mezzanine, etc.) erected to prevent employees and objects from falling to lower levels.

3.13 Hole – a gap or void two inches or more in its least dimension, in a floor, roof, or other walking/working surface.

3.14 Lanyard – a flexible line of rope, wire rope, or strap which generally has a connector at each end for connecting the full body harness to a deceleration device, lifeline, or anchorage.

3.15 Leading Edge – the unprotected side and edge of a floor, roof, formwork for a floor, or deck/platform which changes location as additional surfaces are constructed.

3.16 Lifeline – a component consisting of a flexible line for connection to an anchorage at one end to hang vertically (vertical lifeline), or for connection to anchorages at both ends to stretch horizontally (horizontal lifeline), and which serves as a means for connecting other components of a personal fall arrest system to the anchorage.

3.17 Personal Fall Arrest System (PFAS) – a system used to arrest an employee in a fall from a working level. A PFAS consists of an anchorage, connectors, full body harness, lanyards, deceleration devices, lifelines, or suitable combinations of these.

3.18 Personal Fall Restraint System (PFRS) – means a fall protection system that prevents the user from falling any distance. The system is comprised of an anchorage, a full body harness, connectors, and other equipment such a lanyard or lifeline, etc.

3.19 Qualified Person – a person with an engineering degree and who is knowledgeable in structural construction, design, and the requirements of fall protection systems.

3.20 Safety Nets – a stretched out mesh netting system supported by panels and wires connected to the side of the building designed to catch a worker who has fallen off a leading edge.

3.21 Scaffold – A temporary elevated platform and its supporting structure used for supporting workmen or materials or both.

3.22 Snap hook – a connector comprised of a hook-shaped member with a normally closed keeper, or similar arrangement, which may be opened to permit the hook to receive an object and, when released, automatically closes to retain the object.

3.23 Rescue – the safe recovery of a fallen individual from a hanging position.

3.24 Total Fall Distance – means the sum of the length of lanyard (6ft), the deployment of personal energy absorber (3½ft), the harness stretch (typically 1ft) , the average distance from back D-ring to the sole of the foot on 6’ worker (5ft), and a safety factor (3ft), equals the total fall distance (18 ½ ft).

3.25 Tranfasteners – a star-wheel device which allows continuous movement along a cable and eliminates the need for the user to detach and reattach when changing directions or crossing cable supports.

3.26 Unprotected sides and edges – means any side or edge (except at entrances to points of access) of a walking/working surface (floor, roof, ramp, or runway) where there is no wall or guardrail system at least 39 inches high.

## 4.0 Fall Protection Systems:

4.1 One of the following approved fall protection systems shall be used whenever working in the DCF over four feet from surface to the next level.

4.2 Guardrails – A guardrail system consisting of the following: (1) A top rail at least 42 inches in height above walking level and capable of withstanding a horizontal load of at least 200 pounds; (2) Mid-rails positioned as 21 inches in height and capable of withstanding a horizontal load of at least 150 pounds; (3) And where tools or equipment may be likely to fall from a work surface height to a working level below shall be protected by 4 inch minimal height toe-boards.

4.3 Personal Fall Arrest System (PFAS) – As defined in section 3.0, the PFAS shall (1) only be used by individuals who have been properly trained on BEP's proprietary Mansafe/Latchway Fall Arrest System. The components used as part of the fall arrest systems shall be (2) compatible and designed specifically for fall protection. The BEP's PFAS has anchor points capable of supporting 5,000 pounds for each employee attached. Users of the system shall (3) not exceed the system's weight capacities/forces or stress the system in a manner that may cause possible failure. The total fall distance shall be (4) calculated and approved by a competent person prior to starting work. The system shall be (5) rigged to provide the user with a total fall deceleration distance of no greater than six feet (i.e., the deceleration device shall stop the fall within 3 ½ feet of activation). Components of the system shall be (6) thoroughly inspected by the user and a designated competent person prior to, and at the end of each use. Damaged or worn equipment shall be (7) marked/tagged and not used (i.e., permanently discontinued). An acceptable means of rescue for individuals who may fall shall be (8) provided for prior to being permitted to perform work. Rescue of fallen individuals shall (9) occur immediately if the worker uses an approved device or procedure that would return the worker back to the safety of the working surface, another safe lower surface, or by prompt emergency response of the DC Fire and Rescue Service. Rescue response should occur preferably within 15 minutes of the fall notification/observation. Should a fall occur, the harness, deceleration device, and all harness fittings shall be (10) permanently removed from service.

4.4 Personal Fall Restraint System (PFRS) - this system, as defined in section 3.0, shall be (1) used only when there is no danger of having to arrest a fall. The length of travel of the horizontal lifeline shall be (2) arranged in such a manner that it does not allow the user to reach an unprotected edge in any direction. Only individuals who are properly trained in the use of fall restraint systems shall (3) use the system. System anchor points shall be (4) rated for 3,000 lbs. System components shall be (5) inspected by the user and a designated competent person prior to use. Control line attachment points shall be (6) arranged so as not to present a new hazard (i.e. tripping, etc.). Exact measurements shall be (7) taken prior to being permitted to perform work.

4.5 Safety Net System – As defined in section 3.0, the mesh netting of a safety net system shall be (1) no more than 6" wide. The netting shall be (2) installed by a licensed installer or qualified/competent person and as near the surface as possible but no further than 30 feet below. The installer or qualified/competent person shall (3) test

the system after initial installation and before use, (4) whenever relocated, (5) after major repair, and (6) at 6-month intervals if left in one place. In situations where it is unreasonable to perform the drop test, a qualified/competent person can produce a certification record (e.g., net/installation identification, date of compliance, & signature of qualified/competent person) that the net and net installation has sufficient clearance and impact absorption suitable for use as a fall protection system. The net system shall (6) withstand the force of a 400 pound bag of sand 30" in diameter dropped from the working surface or if not feasible, certify that the net and net installation have sufficient clearance and impact absorption. And the distance from the edge of the working surface to the outer edge of the netting shall be:

| Horizontal And Vertical Safety Net Distances From Work Surface | |
|---|---|
| Net distance below the working surface | Minimum horizontal distance of net from the edge of the working surface |
| Up to 5 feet | 8 feet |
| More than 5 up to 10 feet | 10 feet |
| More than 10 feet | 13 feet |

If a safety net is the appropriate fall arrest solution for the fall hazard present, a detailed site safety plan shall be (7) submitted to OEHS for approval prior to installation. This plan shall (8) contain structural analysis, selection of anchor points, type of netting, calculation of net distances, netting strength (including testing certification of the net and it's components), fall protection associated with safe installation and how to safely rescue workers should a fall occur.

## 5.0 Roles and Responsibilities:

### 5.1 The Office of Environment, Health, and Safety (OEHS) shall:

a. Provide overall technical and administrative management for the program;
b. Develop and update this procedure when necessary;
c. Identify possible fall hazards through regular inspections. Provide a risk assessment/hazard analysis for any identified fall hazard and make appropriate recommendations for protection/abatement;
d. Identify positions requiring fall protection training and ensure those positions are assigned both online and classroom training through ITMS;
e. Ensure written employee fall protection training certification records are in accordance with OSHA requirements;
f. Review renovation, construction, and demolition project safety plans to ensure that appropriate fall protection safeguards are selected as per OSHA and American National Standards Institute (ANSI) guidelines;
g. Review specifications, drawings, and certifications, to determine adequate safety protection (suitable workplace, work height, anchors, wall/floor hole guarding, etc.) is provided and no new hazard is presented.

### 5.2 The Office of Facilities Support (OFS) and Supervisors shall:

a. Immediately address any identified fall protection hazards according to the OEHS hazard analysis/risk assessment;
b. Obtain, install, inspect, test, and maintain fall protection equipment annually and/or as often as necessary;

c. Submit a written safety plan to OEHS for all in-house elevated work projects above 4 feet. The plan shall detail the work process, the proposed fall protection system, the assigned PPE, and detailed communication and rescue procedures. OEHS may request a worksite walk-through meeting to better understand what's being proposed and to assess any additional hazards.
d. Coordinate any contractor instructor-led classroom fall protection training through ITMS.
e. Ensure only trained employees are assigned duties from and/or around elevated work surfaces.
f. Ensure employee and contractor work practices are in compliance with OSHA regulations, approved industry practices, and with this procedure;
g. Ensure employees and contractors don and properly utilize appropriate fall protection equipment;
h. Perform OSHA/ANSI defined competent or qualified person responsibilities as it pertains to installation, inspection, testing/maintenance of fall protection equipment and the systems structural analysis;

**5.3 Contracting officers and their technical representatives shall:**

a. Ensure BEP's fall protection requirements are included in applicable statements of work and project specifications; and
b. Ensure contractors who may work from heights beyond 4 feet submit a site specific safety plan to OEHS through the CO prior to work.

**5.4 Employees and contractors shall:**

a. Attend assigned fall protection training and complete all training objectives/requirements;
b. Only work from heights where adequate fall protection has been installed and only if you've been trained on the fall protection equipment installed;
c. Immediately inform supervisor and OEHS if exposed to areas without adequate fall protection;
d. Inspect fall protection equipment prior to each use;
e. Never use equipment that is frayed or damaged and report findings to supervisor and OEHS;
f. Only use fall protection equipment as the manufacturer designed;
g. Report any falls to their supervisor;

## 6.0 Procedures:

### 6.1 Annex Building Roof (Except Courtyard Roofs)

6.1.1 Workers shall never work alone and shall never access the building roof top without an approved fall protection/rescue safety plan that includes appropriate fall protection equipment in proper working condition.

6.1.2 Workers attempting to gain access to the Annex Building rooftop wings shall notify or have their supervisor notify the Police Communication (874-1234) and the Office of Environment, Health, and Safety (OEHS on 874-2463, 874-3591, or 874-2193) prior to access. Signs shall be posted on roof access doors to reflect this procedure.

6.1.3 Upon proper notification, the OEHS shall assess the nature of work to be performed and the PPE or other protections required prior to granting access.

6.1.4 Only trained employees and contractors who are familiar with BEPs Fall Arrest or Restraint System shall be permitted to access building rooftop access points.

6.1.5 The BEP has a 100% tie-off policy when working within 6' of the roof's leading edge. Nylon Class III full body harnesses with attached "D" rings with no signs of wear or damage are the required PPE for that work.

6.1.6 There are tie-off cables installed between wings on the North and South sides. Engineered window washers "I" bolts are installed as additional tie-off points around the perimeter of the wings.

6.1.7 Although there are no door warning labels, roof access doors are physically locked and secured by Police Communications. Police Communications shall contact OEHS whenever a request is made to open a roof access door.

6.1.8 Penthouse roof is a controlled access zone. Approved access shall only occur through 9th Floor access door # 908 which opens outside to Wing 3, North. Upon careful inspection from a qualified/competent person, workers are to use the external stairs from the wing roof top to gain entry to the penthouse roof . **However, since there is no OSHA complaint fall protection presently installed for that area, access to the penthouse roof is prohibited.**

6.1.9 9th Floor access doors to rooftop Wings 2, 3, 4, and 5 require authorization, PPE, and special fall protection training. Contact the Police Communications and the OEHS to initiate access.

6.1.10 9th Floor access door to rooftop Wing 1 opens to a courtyard with appropriate OSHA compliant permanent guardrails and therefore requires authorization for entry, but shall not require additional PPE and special training (unless the work process requires access beyond the guardrail system to the roof leading edge).

6.2 **Main Building Roof**

6.2.1 Workers shall not work alone on BEP roof tops and shall never access the building roof top without appropriate fall protection equipment in proper working condition.

6.2.2 Workers attempting to gain access to the Main Building roof shall notify Police Communications and the OEHS prior to access. Signs shall be posted on roof access doors to reflect this procedure.

6.2.3 Upon proper notification, the OEHS shall assess the hazards involved with the nature of work to be performed and the PPE required prior to granting access.

6.2.4 Only trained employees and contractors shall be permitted to access building rooftops and to use the BEP Fall Arrest or Fall Restraint Systems.

6.2.4 The BEP has a 100% tie-off policy when working within 6' of the roof's edge. Nylon Class III full body harnesses with attached "D" rings with no signs of wear or damage are the required PPE for that work. Workers are required to use (2) adjustable lanyards with shock absorbing or self retracting nylon lanyards with double locking snap assemblies at each end.

6.2.5 Workers shall attach one lanyard to overhead "I" beam bolts before climbing through windows to access and attach the other lanyard to one (1) of two (2) tranfasteners that is on the cable outside, just above the windows. Additional

tranfasteners may be obtained upon request from the Electric Shop at 874-2164.

6.2.6 The Main Building roof has access through the west side windows of the 5th floor men's room and from room 535-M. The roof can also be accessed by stairs leading to the elevator control room at the 5th Floor Roof Garden.

6.3 **Elevated Work Platforms (e.g., scaffolds, hopper/tank platforms, mobile or stationery lifts, etc.)**

6.3.1 A risk assessment of the working environment shall be carried out by a designated qualified/competent person and controls shall be put in place to reduce the risk of injury or harm when working with all elevated work platforms.

6.3.2 Where the risk of injury or harm to persons cannot be controlled, a PPE assessment shall be carried out and relevant PPE issued.

6.3.3 Scaffolds shall be furnished, erected, or used whenever persons are engaged in work that cannot be performed safely from the ground.

6.3.4 Scaffolds shall be designed and erected to safely support the intended load.

6.3.5 Scaffold footing shall be sound, rigid, and capable of carrying the maximum intended load. Barrels, boxes, brick, concrete block, and other unstable objects shall not be used to support scaffolds or planks. Screw jacks may be used for scaffold leveling on a sound rigid surface. However, not more than 12 inches of the screw jack shall extend below the bottom of the nut/top of caster. Mudsills shall be provided when erecting scaffold on any surface other than concrete.

6.3.6 Anchorage, guying, tying off, or bracing of scaffolds shall be affixed to substantial and structurally sound structures, or the equivalent, using anchor bolts or equivalent.

6.3.7 Guardrails (with complaint top-rails, mid-rails, and toe boards) shall be installed on all open sides and ends of platforms more than 10 feet above the ground or floor or other platform.

6.3.8 Scaffold access shall be provided for every scaffold erected. The maximum spacing between the climbing surfaces of the scaffold frame shall not exceed 16.5" and the length of the climbing surface shall not be less than 10". Scaffold access may also be accomplished by either an internal integrated stair unit or by a hook-on attachment ladder specifically designed for the intended purpose. Ladders shall be positioned so that the scaffold cannot be tipped. Cross braces shall NOT be used as a means of access or degrees.

6.3.9 Scaffold platforms (working surfaces) shall be fully planked or decked. The plank edges shall lay close together so that the platform is tight with no spaces through which tools or fragments of material can fall. Spaces between planks shall NOT be greater than 1" wide. Planks shall lap their end supports at least 12" but not more than 18". Where the ends of the planks form a flush floor, the butt joint shall be at the centerline of a pole and the butt ends shall rest on separate bearers. When two or more scaffolds are used on a building or structure, they shall not be bridged to one another but shall be maintained at an even height with platform butting. Intermediate beams shall be provided where necessary to prevent dislodgement of planks due to deflection and the ends shall be nailed or cleated to prevent dislodgement.

6.3.10 The poles, legs, and uprights of the scaffolds shall be plumb and shall be securely and rigidly braced to prevent swaying and displacement.

6.3.11 Fabricated frame scaffolds including components (i.e., braces, brackets, trusses, screw legs, ladders, etc.) shall be designed to support their own weight and at least 4 times the maximum intended load.

6.3.12 All exposed surfaces shall be free of sharp edges, burrs, nails, or similar safety hazards.

6.3.13 Scaffolds shall not be setup or used in the vicinity of energized electrical lines or conductors unless such are insulated, de-energized (i.e., LO/TO), or are otherwise rendered safe against electrical contact.

6.3.14 Only experienced scaffold installers who are well trained in scaffold construction and fall hazard safety shall be used to erect scaffolds.

6.3.15 Supervisors shall not assign workers tasks they are not trained and competent to perform.

6.3.16 If the hazard assessment or observed work practices determines further training is required such training shall be carried out immediately and/or resources made available prior to being permitted to complete the work task.

## 7.0 Training for Contractors:

Any contractor exposed to fall hazards at BEP must have a written certification record of fall protection training to the Contracting Officer prior to work. Any contractor who will use BEP's Mansafe® Fall Protection System on either the Main or Annex Roof must be trained by a licensed installer such as, Unistrut Distribution Company, 1140 W. Thorndale Ave. Itasca, IL 60143, (800) 468-9510. All training certificates must be sent through the CO to OEHS prior to work. CORs must specify this requirement in applicable SOWs.

## 8.0 Related Documents:

EHS Contractor Requirements 75D-07.0-04
Construction Safety Permits 75D-04.0-08
EHS Review 75D-07.0-01
EHS Records 75D-00.0-12

### 9.0 Document Revision History

| Rev. # | Purpose of Revision | Eff. Date | Approved by: |
|---|---|---|---|
| 1 | Change document numbers to new scheme | 4/15/14 | N. Mohlmann |
| 2 | Three year review and update. | 6/16/17 | N. Mohlmann |
| 3 | Three year review and update | 8/13/20 | V. Antoine-Pompey |
| | | | |